Dylan Becchetti

Mechanical Engineering | Engineering Software & Automation

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Portfolio: formsby-systems.github.io/engineering-portfolio

PROFILE

Lehigh mechanical engineering graduate with three Cardinal Glass manufacturing internships. I build production tools, research and validation software, and parametric mechanical designs. Available January 2027 for full-time U.S. remote work.

EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024–2025 | Engineering Intern, summer 2023

- Built an Excel/VBA tracker that combined separate R&D line files into a daily part-status view used in shift debriefs. Defined the view with engineers and operators.
- Wrote illustrated assembly instructions, worked on spare-parts inventory and layouts, and investigated cleanroom airflow with a hood and micromanometer.
- Inspected pilot-line equipment, analyzed data in Minitab, drafted AutoCAD layouts and contributed to design failure mode and effects analysis.

SELECTED PROJECTS

Midas Futures | Creator & Technical Lead

- Led a year of daily development on a Python platform with pluggable strategies, deterministic batch replay, checkpointed ablations, Monte Carlo validation and controlled broker Practice execution.
- Unified historical and forward decision rules, matching a sealed reference on 228 sessions and 225 modeled trade lifecycles. Used 200,000 selection-aware draws to test a separate historical result.

Wilbur Powerhouse | Mechanical Design, year-long team capstone

- Led major design revisions and modeled the full-scale suspended sculpture in OpenSCAD, including a 79.5-inch ring gear, welded steel carrier and wheel interfaces. Worked with the team on carrier FEA, CNC prototyping and fit checks.

Operalith | Founder & Product Lead, independent project

- Built a Next.js workspace linking tasks, projects and supplies across five tools. Designed PostgreSQL saves that return prior results on retry and reject stale edits. Verified an edit survived reload and undo preserved 11 prior history entries.

Scanned Leverage and Optimized Production | Creator

- Built a Python pipeline that compiles different Roblox place files from the same hash-pinned Luau modules. Connected market discovery to reusable components and CI-checked build recipes.

Formsby | Creator, search prototype

- Designed a cross-platform request router and built the housing prototype with queued searches, listing verification and a private results dashboard.

EDUCATION & TOOLS

Lehigh University | B.S. Mechanical Engineering, May 2026

Materials Science and Engineering minor expected December 2026. SOLIDWORKS CAD Design Associate (CSWA), Dec. 2024.

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Minitab.